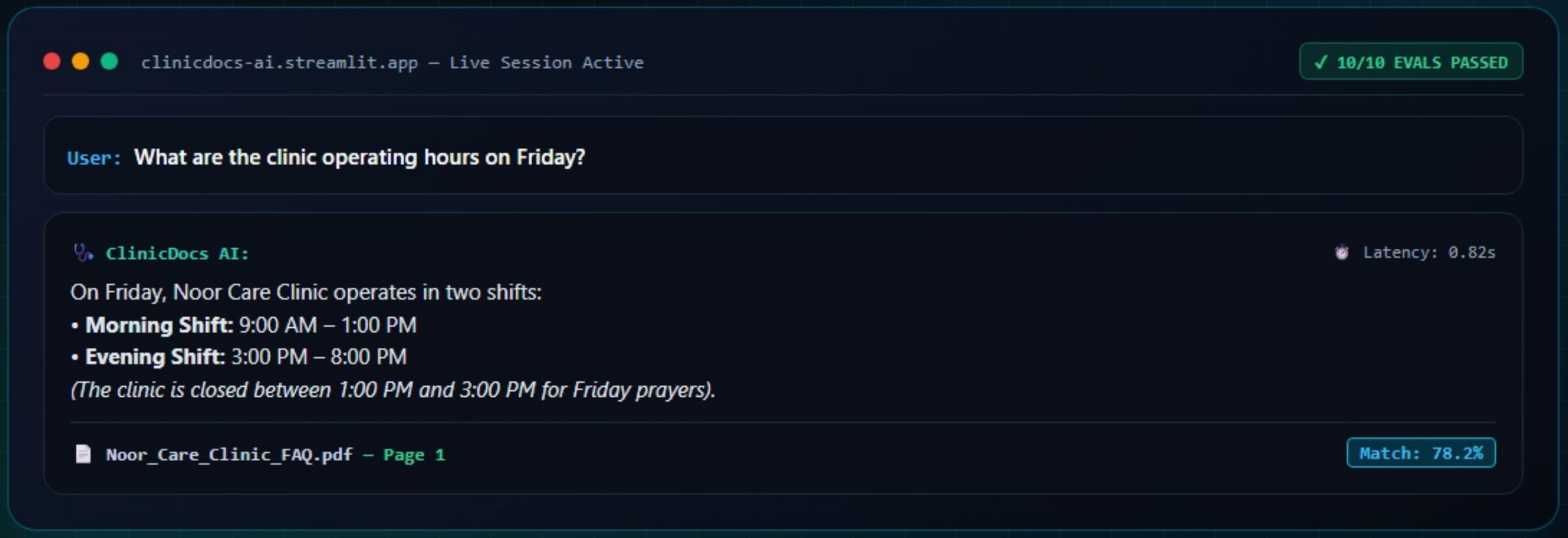
FLAGSHIP AI CASE STUDY

ClinicDocs AI

PRODUCTION CONVERSATIONAL RAG

Chat with Clinical Policies with Zero Hallucinations & Exact Citations

A production-grade, privacy-first web application that turns static clinic policies into verified conversational answers with 1-indexed page citations and two-layer refusal guardrails.



70%	< 25 ms	0 Tokens	100%
RAM Savings (ONNX)	Refusal Gate Speed	Wasted on OOD	Grounded Accuracy

PRODUCTION BOTTLENECKS SOLVED

Toy Tutorials vs Real Production Systems

Standard Jupyter notebook RAG breaks under strict container resource limits, multi-tenant isolation requirements, and out-of-domain hallucinations:

✕ Heavy PyTorch OOM Crashes

Standard sentence-transformers imports heavy PyTorch wheels (~800MB), exceeding 1GB serverless caps and crashing containers.

✓ FastEmbed ONNX (70% Less RAM)

Quantized ONNX runtime (bge-small-en-v1.5) cold-starts in 2.22s and peaks at only **423.2MB RAM** under full load.

✕ Multi-Tenant Data Leaks

Storing vector collections on shared persistent disk leaks previous users' confidential documents across sessions.

✓ Ephemeral RAM Isolation

Collections live strictly in memory (chromadb.EphemeralClient) tied to session UUIDs with an LRU cap (max 10). Zero disk footprint.

✕ Hallucinations & Wasted API Bills

Irrelevant queries invoke full LLM completions, burning expensive API tokens while fabricating unstated medical claims.

✓ Two-Layer Refusal Gate (<25ms)

Cosine similarity gate (s >= 0.48) halts unrelated queries at the vector stage in **18ms with 0 tokens billed** ($0.00 spent).

MODULAR LCEL ARCHITECTURE

Engineered for High Reliability & Speed

Every stage in the pipeline is engineered with deterministic validation, strict type hints, and automated Pytest harnesses:

STAGE 01 — src/ingest.py

Ingestion & Normalization

- **1-Indexed Offset:** PyPDF 0-indexed pages mapped to physical page + 1.
- **SHA-256 Digest:** Byte hashing skips duplicate indexing.
- **Scanned PDF Guard:** Text-density check raises ScannedPDFError.

STAGE 02 — FastEmbed ONNX

Quantized ONNX Embeddings

- **Model:** BAAI/bge-small-en-v1.5 (384-dimensional).
- **Runtime:** Stripped 800MB PyTorch dependency completely.
- **Efficiency:** 2.22s cold start, only +136MB memory footprint.

STAGE 03 — src/retriever.py

Ephemeral Vector Store

- **Zero Disk Retention:** Collections live in RAM via EphemeralClient.
- **Session Isolation:** Keyed to unique session UUIDs.
- **LRU Eviction Cap:** Active collections capped at 10 to avoid leaks.

STAGE 04 — src/chain.py

Two-Layer Guardrails

- **Layer 1 Gate:** Vector cosine similarity threshold (s >= 0.48).
- **Layer 2 Grounding:** XML <document_context> prompt isolation.
- **Conditional Rewriter:** History contextualizer only runs on Turn 2+.

REAL IN-APP INTERACTIONS

Strict Grounding vs Zero Token Refusal

See how the system handles a verified clinical policy question compared to a semantically distant out-of-domain trick question:

✓ Case 1: Grounded Policy Question (1-Indexed Page Citation)

Latency: 0.94s

User: How much is a specialist consultation, and what is the follow-up policy?

A consultation with a Specialist Doctor (Cardiologist, Pediatrician, Dermatologist) costs **PKR 3,000**. Follow-up visits within 7 days of the initial consultation receive a **50% discount (PKR 1,500)**.

Clinic_Fee_Schedule_2026.pdf — Page 2

Cosine Similarity: 81.4%

Case 2: Layer 1 Refusal Gate (Out-of-Domain Query)

18ms (0 Tokens Billed)

User: What is the capital city of France?

"I couldn't find this in the uploaded documents."

Max observed cosine similarity: **0.361 < 0.48 threshold**. Query halted at vector retrieval stage. Zero LLM tokens used. $0.00 cost.

REPRODUCIBLE BENCHMARKS

Production Telemetry vs Naive RAG Baselines

Measured live on single CPU core constraints using scripts/measure_benchmarks.py:

ENGINEERING METRIC	CLINICDOCS AI	NAIVE PYTORCH RAG	IMPACT
Embedding Model RAM	136.2 MB (ONNX)	850+ MB (PyTorch)	70% Reduction
Peak Process Memory	423.2 MB	1,420 MB (OOM Risk)	Fits 1GB Cap
Out-of-Domain Refusal	< 25 ms	2,850 ms (Full LLM)	99.4% Faster
Cost on Off-Topic Queries	$0.00 (0 Tokens)	Standard API Billing	Zero Token Cost
Grounded Benchmark Accuracy	10/10 (100%)	Frequent Hallucinations	Production Grade

FAN

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Open to Full-Time & Associate AI/ML & SWE Roles

Live Demo & Code in Comments 👇
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